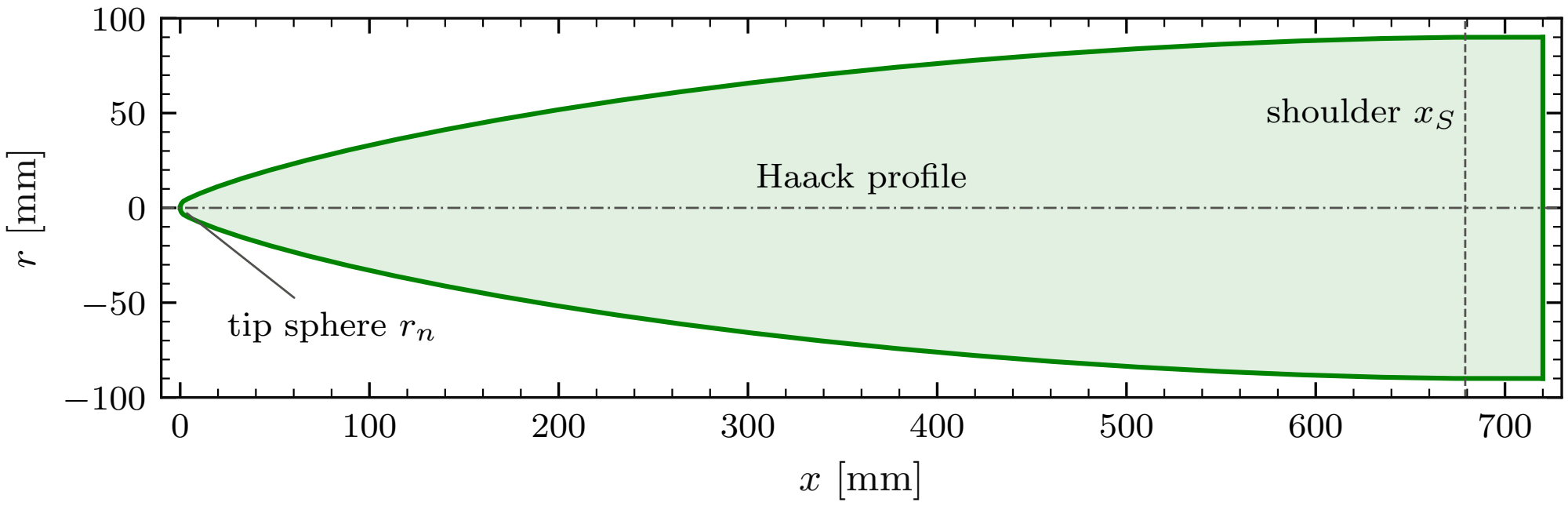

100
50
0
−50
−100
r [mm]
0
100
200
300
400
500
600
700
x [mm]
shoulder x_S
Haack profile
tip sphere r_n